

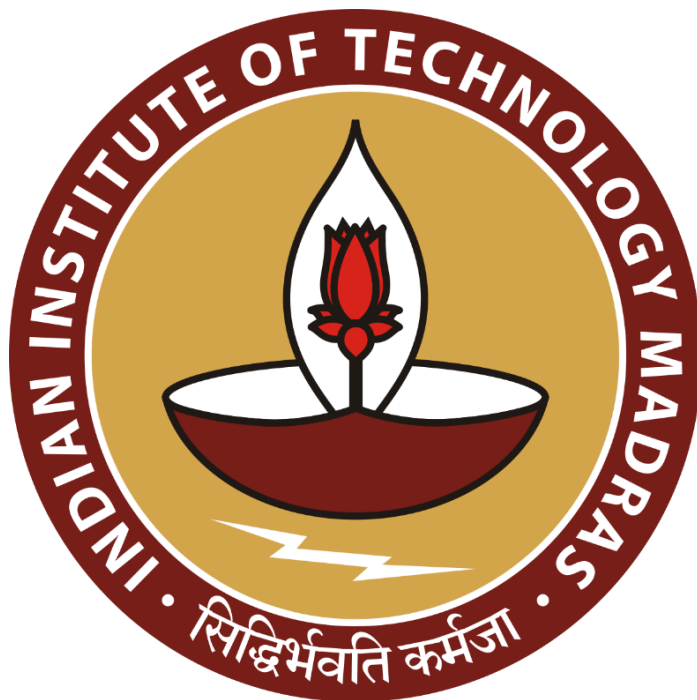
Business Data Management for a Dual Market Utility Goods Store: A Case Study on Jhansi Rope Store

A Proposal Report for BDM Capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “ **Business Data Management for a Dual Market Utility Goods Store: A Case Study on Jhansi Rope Store** ”. I extend my appreciation to **Jhansi Rope Store**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.



Signature of Candidate: (**Digital Signature**)

Name: Pooja Sahu

Date: 10/07/25

1. Executive Summary and Title

Project Titled: “Business Data Management for a Dual Market Utility Goods Store: A Case Study on Jhansi Rope Store”

This proposal presents a data-driven approach to solving key inventory and demand forecasting issues at *Jhansi Rope Store*, a third-generation, family-run utility goods store located in Subhash Ganj, Jhansi. The store operates in both B2B and B2C markets, supplying ropes, niwar, mops, wipers, and related household items to urban and rural customers.

The business faces several operational challenges, including stockouts, overstocking, seasonal demand fluctuations, and an inability to identify fast- and slow-moving products. These problems lead to blocked working capital and missed revenue opportunities.

To address these challenges, historical sales data will be analyzed to uncover patterns in product movement and seasonal demand. Techniques such as descriptive analytics, trend analysis, and demand forecasting will be applied using tools like Microsoft Excel for initial data aggregation and visualization, and Python (Pandas, Matplotlib, and Scikit-learn) for deeper statistical and predictive analysis.

By integrating historical data with analytical methods, the project aims to develop a smart inventory management strategy. The expected outcome is a reduction in stock-related inefficiencies, better product availability, improved cash flow, and enhanced profitability, helping the business make more informed stocking decisions going forward.

2. Organizational Background

Established in 1965, Jhansi Rope Store is a third-generation, family-run business located in Subhash Ganj, Jhansi. The store has built a strong reputation as a reliable source for everyday utility products such as mops, wipers, ropes, niwar, and twine. It serves both retail and wholesale customers across urban and rural areas, including households, small retailers, farmers, and service providers. Now managed by Anshu Sahu, the founder’s grandson, the store continues to uphold a legacy of trust, quality, and community connection. By sourcing products directly from manufacturers, it ensures affordability, consistent supply, and timely availability. The business generates an average monthly revenue of around ₹1 lakh, with an annual turnover of approximately ₹1.5 crore, supported by steady demand and seasonal peaks. From its roots as a small neighborhood shop, Jhansi Rope Store has grown into a well-established local enterprise known for its practical service and long-standing customer relationships.

3. Problem Statement

Jhansi Rope Store struggles with ineffective inventory and demand forecasting due to the absence of data-driven systems, leading to frequent stock imbalances, poor planning, and reduced operational efficiency.

3.1 Inability to Forecast and Adapt to Seasonal and External Demand

Due to the lack of historical sales analysis and trend tracking, the store fails to anticipate seasonal spikes or dips in demand. Additionally, external factors such as festivals, weather changes, and urban–rural market variations are not accounted for, resulting in overstocking or stockouts during peak periods.

3.2 Lack of Clarity on Product Movement

The absence of proper data analytics makes it difficult to distinguish between fast- and slow-moving products. This leads to inefficient stocking decisions, with capital getting locked in unsold inventory while demand for high-moving items often goes unmet.

3.3 Manual Inventory Management Systems

The current reliance on manual processes limits visibility into real-time stock levels and slows down the restocking process. This hampers the store’s responsiveness to market demand and increases the chances of inventory mismatches.

4. Background of Problem

Inability to forecast and adapt to seasonal and external demand: Jhansi Rope Store currently lacks historical sales analysis and trend tracking mechanisms. As a result, the store is unable to accurately predict seasonal changes in demand, such as during festivals, agricultural cycles, or academic year openings. Moreover, external factors like weather fluctuations and the differing needs of rural and urban customers are not integrated into planning. This often results in overstocking or stockouts during peak periods, directly impacting sales performance and customer satisfaction.

Lack of clarity on fast- and slow-moving products: Without proper data tracking, the store is unable to identify which products sell quickly and which remain unsold for long periods. This creates challenges in optimizing shelf space and restocking decisions. Slow-moving items block working capital and storage space, while fast-moving items may run out when needed most. This imbalance contributes to overall inefficiency in operations.

Manual and inefficient inventory tracking system: The current inventory management process relies heavily on manual records and staff memory, which is prone to human error. This results in outdated stock counts, delayed restocking, and inconsistent supply. Lack of automation also makes it difficult to get a clear, real-time view of stock availability. These inefficiencies hinder the store’s ability to respond quickly to customer demand.

5. Problem Solving Approach

To tackle the inventory and demand forecasting challenges faced by Jhansi Rope Store, a structured, data-driven problem-solving approach was implemented. This approach is divided into three key components: **Data Collection**, **Analytical Methods**, and **Tools Used**—each contributing to the generation of actionable business insights.

5.1 Data Collection

Primary data was sourced from the store's manual sales registers, capturing product name, date of sale, quantity sold, unit price, and customer type (retail or wholesale). The data covers periods with known seasonal variation, such as festivals, agricultural cycles, and school reopenings. Inventory records—stock received and stock remaining—were also included to identify patterns of overstocking and stockouts.

5.2 Analytical Methods

The data was first cleaned to correct errors, fill missing values, and standardize formats, then structured using spreadsheets or a relational database.

Descriptive analysis was applied to:

- Identify seasonal and monthly sales trends
- Detect fast- and slow-moving products
- Uncover recurring inventory mismatches

A contextual analysis was also conducted to explore correlations between external factors (e.g., festivals, weather, rural vs. urban buyers) and changes in demand patterns.

5.3 Tools Used

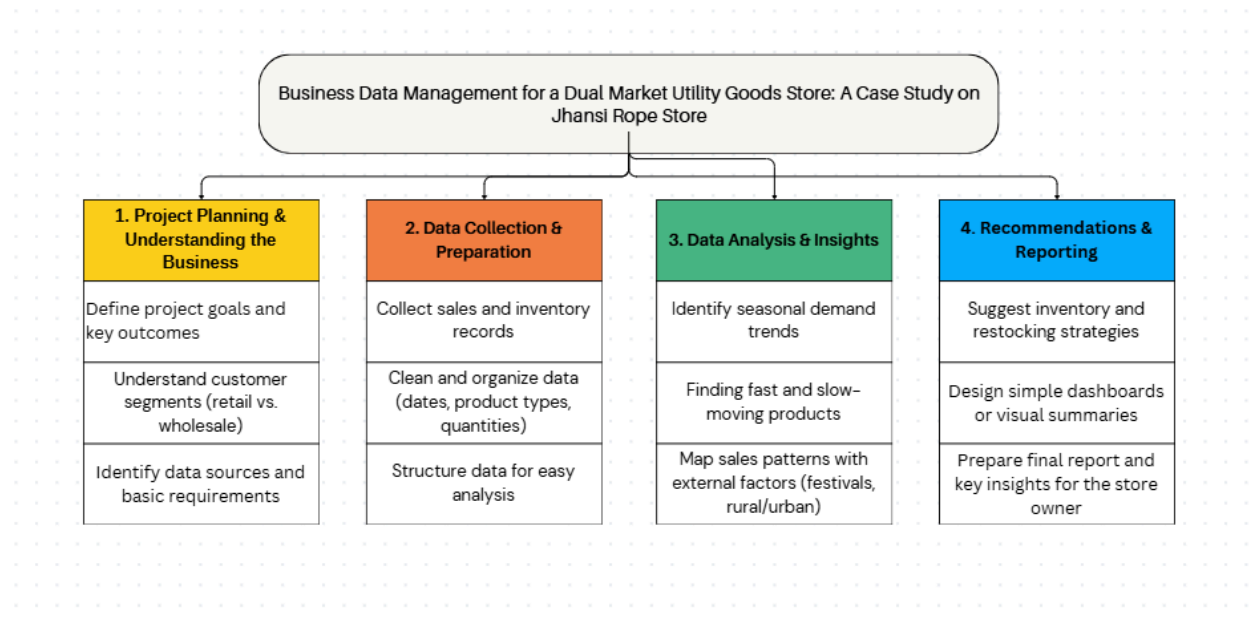
A combination of tools supported the analysis:

- **Microsoft Excel**: Used for cleaning, organizing, and basic analysis via PivotTables and charts
- **Python (Pandas, Matplotlib, Seaborn)**: Enabled in-depth analysis and visualizations such as trend lines and heatmaps.
- **R (optional)**: Provided statistical summaries and evaluation of external factors
- **Google Sheets**: Allowed cloud-based access and collaboration with store personnel

These tools collectively enabled a comprehensive understanding of sales patterns and inventory flow, supporting better business decisions.

6. Expected Timeline

6.1 Work Breakdown Structure



The project will follow the steps shown below, organized under three key submission phases:

Proposal Phase (11–13 July 2025)

- Identify project topic and relevance
- Draft and submit project proposal (*submitted on 11 July*)
- Proposal approval and feedback incorporation (*Mid–Late July*)

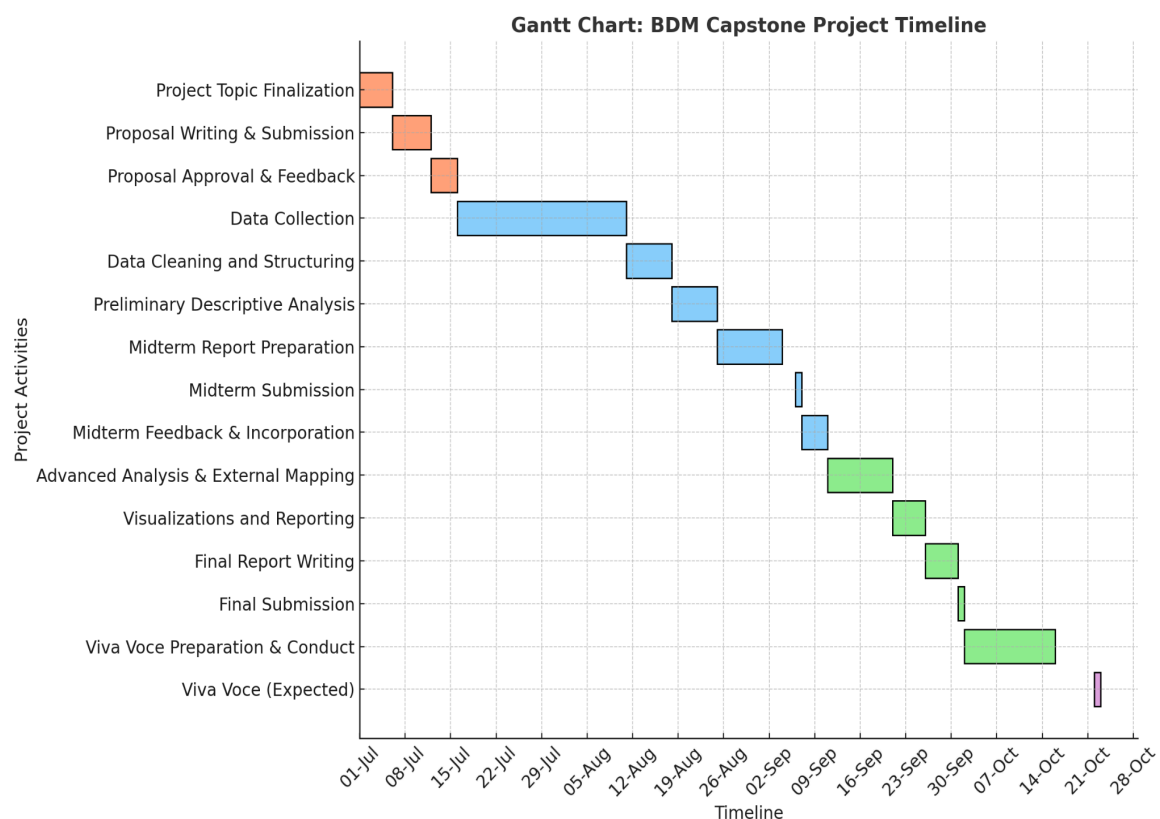
◆ **Midterm Phase** (6–8 September 2025)

- Data collection from manual sales records (*July–August*)
- Data cleaning and formatting (*Late August–Early September*)
- Preliminary descriptive analysis (*Early September*)
- Midterm report preparation and submission (*within 6–8 September*)
- Incorporation of midterm feedback (*Mid–September*)

◆ **Final Phase** (1–3 October 2025)

- Advanced data analysis and interpretation (*Mid–Late September*)
- Mapping of external factors (seasonality, rural-urban trends, etc.)
- Visualizations using Excel/Python/R (charts, pivot tables, etc.)
- Final report preparation and submission (*within 1–3 October*)
- Final approval (*Early–Mid October*)
- Viva Voce (*Post 22 October 2025*)

6.2 Gantt Chart



7. Expected Outcomes

1. **Clear Seasonal Demand Insights**

By analyzing historical sales data across different time periods, the project will uncover recurring **seasonal patterns**—especially around festivals, harvest seasons, and monsoon periods. These insights will support the development of basic **forecasting models** (e.g., moving averages, seasonal decomposition) to help the store plan stock purchases in advance, reducing both overstocking and stockouts.

2. **Visibility into Product Movement**

The project will analyze product-level sales to identify **fast-moving and slow-moving items**. Using **ABC analysis**, products will be categorized based on their contribution to revenue or sales frequency. This will help the store prioritize procurement, allocate shelf space efficiently, and reduce storage of stagnant inventory.

3. **Organized and Efficient Inventory Tracking**

By replacing manual inventory tracking with structured spreadsheets or simple inventory tools, the store will gain a **more organized and reliable system**. This will improve the accuracy of stock records, ensure timely restocking, and help prevent inventory-related losses.

4. **Data-Driven Planning for External Factors**

The analysis will explore how **weather patterns, customer type (rural vs. urban), and local events** influence demand. To plan better for these external factors, the project will use:

- **Regression analysis** to quantify the impact of variables like rainfall or temperature on product demand.
- **Customer segmentation** to compare purchase trends between rural and urban buyers.
- **Event impact analysis** to examine sales data around festivals and local markets.

5. These methods will enable the store to develop **scenario-based stocking strategies**, improving preparedness for fluctuations caused by external events.

6. **Practical and Actionable Recommendations**

The final report will provide **easy-to-use visual dashboards** and charts using tools like **Excel or Power BI**, along with clear, actionable insights. These tools will support the store in making better inventory and demand-related decisions across both its **B2B and B2C operations**, without the need for complex software.